

Chapter-THREE

Part-II

Risk_Return Analsys for Portfolio

Learning Objectives

1. Understand the concept of portfolio
2. Undertake Portfolio risk-return analysis of two or more securities
3. Discuss the concepts of correlation and diversification

Introduction

- **Risk–return** trade-offs have an important role to play in corporate **finance theory** – from both a company and an investor perspective.
- Companies face variability in their **projected cash flows** whereas investors face variability in their **capital gains and dividends**.
- Assuming that companies and shareholders are rational, their aim will be to minimise the risk they face for a given return they expect to receive.
- So in this chapter we will examine how investors, by **‘not putting all their eggs in one basket’**, are able to reduce the risk they face given the level of their expected return.

Con't...

- **PORTFOLIO**: is a collection or a group of investment assets. If you hold only one asset, you suffer a loss if the return turns out to be very low.
- If you hold two assets, the chance of suffering a loss is reduced and returns on both assets must be low for you to suffer a loss.
- By diversifying, or investing in multiple assets that do not move proportionately in the same direction at the same time, you **reduce your risk**.

Con't...

- **Portfolio analysis** is used to determine the **return and risk** for these combinations of assets.
- It is the **total portfolio risk and return** that is important.
- The risk and return of individual assets should not be analyzed in isolation; rather they should be analyzed in terms of how they affect the **risk and return of the portfolio** in which they are included.
- The goal of the financial manager should be to create an efficient portfolio, one that **maximizes return for a given level of risk or minimizes risk for a given level of return.**

Portfolio Theory

Two Approaches to Constructing Portfolios

I. Traditional Approach

versus

II. Modern Portfolio Theory

Traditional Portfolio Theory

- An investment advisor would have given you advice like:
 - *If you are young you should be putting money into a couple of good **growth stocks**, maybe even into a few small stocks.*
 - *If you're close to retirement, you should be putting all of your money into **bonds and safe stocks**, and nothing into the risky stocks*
- Tends to focus on well-known companies
 - Perceived as less risky
 - Stocks are more liquid and available
 - Familiarity provides higher “comfort” levels for investors

Modern Portfolio Theory (MPT)

- It deals with the selection of **optimal portfolios** by rational risk averse investors-i,e by investors who attempt to maximize their expected portfolio return consistent with individually acceptable levels of portfolio risk.
 - Given a choice between two assets with equal rates of return, risk-averse investors will select the asset with the lower level of risk.
- **A risk-averse investor will prefer**
 - a portfolio with the highest expected return for a given level of risk or
 - prefer a portfolio with the lowest level of risk for a given level of expected return. In portfolio theory, this is referred to as the *principle of dominance*.
- Portfolio theory determines the combination of risk and return that allows the investors to achieve the highest return for a given level of risk.

Modern Portfolio Theory (MPT)

- Emphasizes statistical measures to develop a portfolio plan
- **Focus is on:**
 - Expected returns
 - Standard deviation of returns
 - Correlation between returns

Markowitz Portfolio Theory

■ Quantify risk

- ✂ In the early 1960s, the investment community talked about **risk of the portfolio**, but there was no specific measure for the risk of the portfolio.
- ✂ To build a portfolio model, however, investors had to quantify their **risk variable**.
- ✂ **Portfolio theory** , originally proposed by **henry markowitz** in the 1952, was the first formal attempt to quantify the risk of a portfolio and develop a methodology for determining the optimal portfolio.
 - He was the first person to show **quantitatively why and how diversification reduces risk**.
 - Derives the expected rate of return for a portfolio of assets and an expected risk measure
 - Derives the formula for computing the variance of a portfolio, showing how to effectively diversify a portfolio

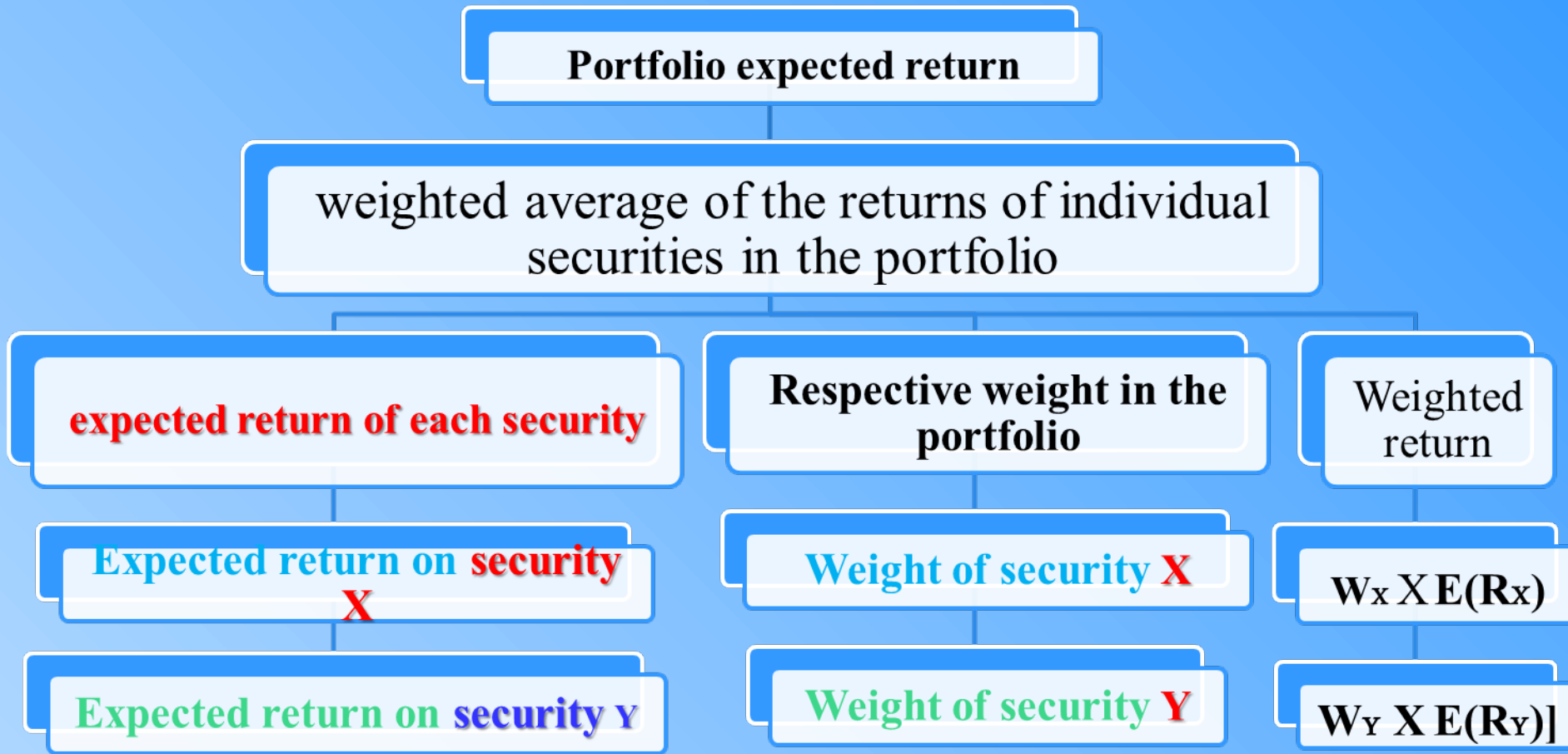
PORTFOLIO RETURN: TWO-ASSET CASE

↗ The return of a portfolio is equal to the **weighted average of the returns of individual assets** (or securities) in the portfolio.

↗ Requirement

- Expected return of individual security
- Portfolio weight
 - Portfolio weights: proportion of fund invested in each security / Percentages invested in each asset (w_i) serve as the weights
- Multiply by respective weight and sum the result

Expected return of a portfolio



$$E[r] = P_1 r_1 + P_2 r_2 + P_3 r_3 + \dots P_n r_n$$
$$= \sum_{i=1}^n P_i r_i$$

$$E(R_p) = [W_X \times E(R_X)] + [W_Y \times E(R_Y)] + \dots [W_n \times E(R_n)]$$

$$E(R_p) = \sum_{i=1}^n W_i E(R_i)$$

Con't....

- The **expected return** on a portfolio, $\hat{r}_p$, is simply the weighted average of the expected returns on the individual assets in the portfolio, with the weights being the fraction of the total portfolio invested in each asset:

$$\hat{r}_p = w_1\hat{r}_1 + w_2\hat{r}_2 + \dots + w_n\hat{r}_n$$

Formula : $E(R_p) = \sum_{i=1}^n W_i R_i$

Where:

- ✓ R_p = Expected Return of the Portfolio.
- ✓ W_i = The weights attached for each security.
- ✓ $\hat{r}_1, \hat{r}_2, \hat{r}_3, \dots, \hat{r}_n$ = Expected return of a security.
- ✓ n = No of stocks/ securities in a portfolio.

Con't...

- Note; “ W_i ” is the fraction of the portfolio’s dollar value invested in stock/security i , that is, the value of investment in stock i divided by the total value of the portfolio. Thus, the sum of W_i ’s must be 1.00.
- **Example:** Consider a portfolio of three stocks A, B, and C, with expected returns of 16%, 12%, and 20% respectively. The portfolio consists of 50% stock A, 25% stock B, and 25% stock C.
- **Required:** what is the expected return on this portfolio?
- $E(R_p) = .16*50 + .12*25 + .20*25 = 16\%$ which is the expected return of a portfolio

Con't...

Suppose we have the following projections on three stocks:

<i>State of Economy</i>	<i>Probabilities</i>	<i>Returns if state occurs</i>		
		<i>Stock A</i>	<i>Stock B</i>	<i>Stock C</i>
<i>Boom</i>	<i>0.40</i>	<i>10%</i>	<i>15%</i>	<i>20%</i>
<i>Bust</i>	<i>0.6</i>	<i>8%</i>	<i>4%</i>	<i>0%</i>

We want to calculate portfolio expected returns in two cases. First, what would be the expected return on a portfolio with equal amounts invested in each of the three stocks? Second, what would be the expected return if half of the portfolio were in A, with the remainder equally divided between B and C? (8pt)

Another Example

- Consider the following information

– State	Probability	X	Z
– Boom	.25	15%	10%
– Normal	.60	10%	9%
– Recession	.15	5%	10%

- What is the expected return and standard deviation for a portfolio with an investment of \$6000 in asset X and \$4000 in asset Y?

PORTFOLIO EXPECTED RETURN Con't...

- **Example-2:** A portfolio consists of four securities with expected returns of 12%, 15%, 18%, and 20% respectively. The proportions of portfolio value invested in these securities are 0.2, 0.3, 0.3, and 0.20 respectively.

$$E(R_p) = \sum_{i=1}^n w_i E(R_i)$$

where $E(R_p)$ = expected portfolio return

w_i = weight assigned to security i

$E(R_i)$ = expected return on security i

n = number of securities in the portfolio

Then expected return on the portfolio of the above Example is:

$$\begin{aligned} E(R_p) &= 0.2(12\%) + 0.3(15\%) + 0.3(18\%) + 0.2(20\%) \\ &= \mathbf{16.3\%} \end{aligned}$$

Ex Ante Portfolio Returns

Simply the Weighted Average of Expected Returns

Example-3:

	Relative Weight	Expected Return	Weighted Return
Stock X	0.400	8.0%	0.03
Stock Y	0.350	15.0%	0.05
Stock Z	0.250	25.0%	0.06
Expected Portfolio Return =			<u><u>14.70%</u></u>

Risk in a Portfolio Context

- **As** we just saw, the expected return on a portfolio is simply the weighted average of the expected returns on the individual assets in the portfolio.
- However, unlike returns, the risk of a portfolio, **standard deviation of a portfolio**, is generally *not* the weighted average of the standard deviations of the individual assets in the portfolio; the portfolio's risk will almost always be *smaller* than the weighted average of the assets' S.D's.
- Markowitz showed how the risk of a portfolio is **NOT** just the weighted average sum of the risks of the individual securities...but rather, also a function of the degree of co movement of the returns of those individual assets.

- **Portfolio risk** can be reduced by the simple kind of diversification, that is; when different assets are added to the portfolio, the **total risk tends to decrease**.
- **Total risk of portfolio** consists of **systematic risk & unsystematic risk** and this total risk is measured by the **variance** and **standard deviation** of the rate of return overtime.
- As the **portfolio size increases**, the **total risk decreases up to a certain** level. Beyond that limit, risk cannot be reduced.
 - This indicates that spreading out the assets beyond certain level can't be expected to reduce the portfolio's total risk below the level of **un-diversifiable risk**.
- Markowitz showed that the **variance of the rate of return** was a meaningful measure of portfolio risk under a reasonable set of **assumptions**, and he derived the formula for computing the **variance of a portfolio**.

Assumptions of the Markowitz Model:

- **The Markowitz model is based on several assumptions regarding investor behaviour:**
 1. Investors consider each investment alternative as being represented by a **probability distribution of expected returns** over some holding period.
 2. Investors maximize one-period expected utility, and their utility curves demonstrate diminishing marginal utility of wealth.
 3. Investors estimate the **risk of the portfolio on the basis of the variability of expected returns**.

Con't...

4. Investors base decisions solely on **expected return and risk**, so their utility curves are a function of expected return and the expected variance (or standard deviation) of returns only.
5. For a given **risk level**, investors prefer **higher returns to lower returns**. Similarly, for a given level of expected return, investors prefer less risk to more risk.

Con't...

- **Therefore:**

- ✓ Under these assumptions, a single asset or portfolio of assets is considered to be efficient if **no** other asset or portfolio of assets offers **higher expected return** with the same (or lower) risk, or lower risk with the **same (or higher) expected return**.

Diversifying unsystematic risk using a two-share portfolio

- The simplest portfolio to consider is that **containing two shares**.
- ***Diversification*** is the combination of assets whose returns do not vary with one another in the **same direction at the same time**.
- The extent to which a two-share portfolio will reduce unsystematic risk depends on the **correlation** between the two shares' returns.
- This correlation is quantified by the **correlation coefficient (ρ)** of the returns of the two shares, which can take any value in the range **-1 to +1**.

Correlation

- **Correlation (co-movement)**: refers to the association of movement between **two numbers**.
- It measures the **degree** of linear relationship to which two variables, such as **returns on two assets, move together**.
- Correlation takes on numerical values that range from **+1 to -1**.

Con't...

- While the **positive or negative sign** indicates the **direction of the co-movement**.
- The closer the correlation coefficient is to +1 or -1, the stronger the association.
 - +1 is perfect positive correlation.
 - -1 is perfect negative correlation.
 - 0.0 no relationship
- If the variables move together, they are positively correlated (a positive correlation coefficient).
- If the variables move in opposite direction, they are negatively correlated (a negative correlation).

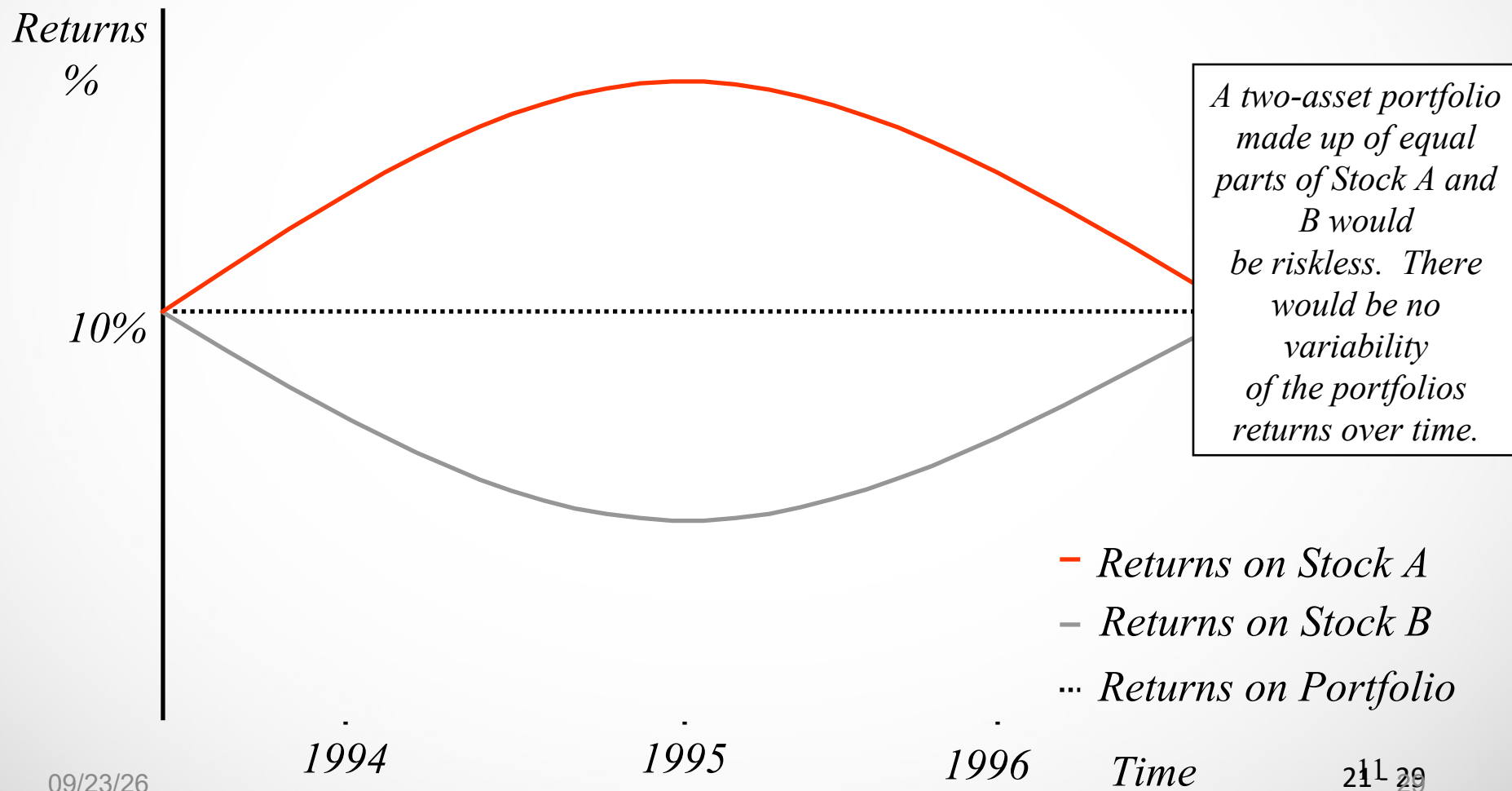
Con't...

- A correlation of **+1.0** indicates that the variables **move up** and **down together**, the relative magnitude of the movements is exactly the same (perfect positive correlation).
- A correlation of **-1.0** implies that they move exactly opposite to each other. (Perfect negative correlation).
- A correlation of **0.0** indicates that **no relationship** between the variables, that is they are unrelated.

Con't...

- If correlation is between 0.0 and +1.0, the returns usually **move up and down together**, but not all the time. The closer the correlation is to 0.0, the lesser the two sets of returns **move together**.
- A **correlation coefficient falling between -1 and 0** indicates negative, but not perfect negative correlation between the two assets' returns.

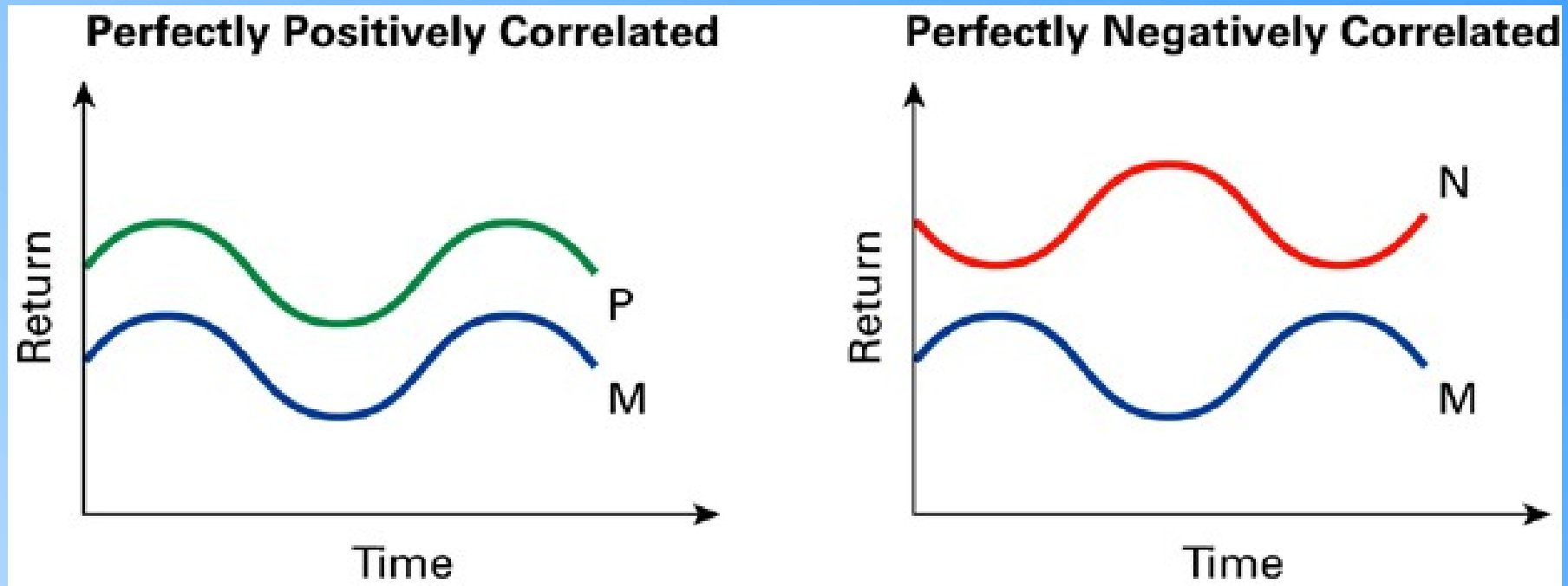
Perfect Negatively Correlated Returns over Time



Con't...

- ✓ If $\rho(A,B) = +1$ no unsystematic risk can be diversified away
 - ✓ If $\rho(A,B) = -1$ all unsystematic risk will be diversified away
 - ✓ If $\rho(A,B) = 0$ no correlation between the two securities' returns
-
- Therefore, when picking a two-share portfolio it is most beneficial to choose two shares whose **correlation coefficient** is as close to **-1** as possible.
 - However, as long as the correlation coefficient is **less than +1**, some unsystematic risk will be **diversified away**.

The Correlation Between Series M, N, and P



Correlation:

Why Diversification Works!

- To reduce overall risk in a portfolio, it is best to combine assets that have a *negative (or low-positive) correlation*.
- *Uncorrelated* assets reduce risk somewhat, but not as effectively as *combining negatively correlated assets*.
- When correlation coefficient of returns on individual securities is perfectly positive (i.e., $\text{cor} = 1.0$), then there is no advantage of diversification.
 - We may therefore conclude that diversification always reduces risk provided the correlation coefficient is less than 1.

Con't...

- In practice it is difficult to find two securities whose correlation coefficient is exactly **-1**, but the most commonly quoted example is that of an **umbrella manufacturer** and **an ice cream company**.
- The classical **example** of the benefit of diversification is to consider the effect of combining investment in **an ice-cream producer** with the investment in a **manufacturer of umbrella**.

Con't...

- For simplicity assume that the return to the ice-cream producer is **+15% if the weather is sunny** and **-10% if it is rain**. Similarly the manufacturer of umbrella benefits when it **rains (+15)** and **looses when the sun shines (-10%)**. Further assume that each of the two weather states occur with probability of 50%.
- *Thus, **diversification** does nothing to reduce risk if the portfolio consists of perfectly positively correlated stocks.*

Co-efficient of Correlation

$$\begin{aligned}\text{Cor } (R_i, R_j) \text{ or } \rho_{ij} &= \frac{\text{Cov } (R_i, R_j)}{\sigma_i \sigma_j} \\ &= \frac{\sigma_{ij}}{\sigma_i \sigma_j} \\ \sigma_{ij} &= \rho_{ij} \cdot \sigma_i \cdot \sigma_j\end{aligned}$$

where ρ_{ij} = correlation coefficient between the returns on securities i and j

σ_{ij} = covariance between the returns on securities
 i and j

σ_i, σ_j = standard deviation of the returns on securities

CO-VARIANCE OF RETURNS

- **Covariance** measures how closely security returns move together. The covariance between possible returns for securities J and K,
- When we consider two assets in the portfolio, we are concerned with the **co-movement** of security movement.
- **Co-movements** between the returns of securities are measured by **covariance (an absolute measure)** and **coefficient of correlation (a relative measure)**.
- It can be:
 - Positive Covariance,
 - Negative Covariance, and
 - Zero Covariance,

Con't...

- A **positive covariance** between the returns of two securities indicates that the returns of the two securities tend to move in the **same direction**.
- A **negative covariance** between the returns of two securities indicates that the returns of the two securities tend to move in **opposite directions**.
- A relatively **small or zero covariance** between the returns of two securities indicates that there is little or no relationship between the returns of the two securities;
- We denote the **covariance** between the return of security **i** and the return of security **j** by σ_{ij} (the Greek letter **sigma**, **σ_{ij}** or covariance **ij**).

COVARIANCE

- **Covariance will flow with the following steps:**
- **Step 1:** Calculate the expected return for each assets
- **Step 2:** For each scenario and investment, subtract the investment's expected return from its possible outcome.
- **Step 3:** For each scenario, multiply the deviations for the two investments.
- **Step 4:** Weight this product by the scenario's probability.
- **Step 5:** Sum these weighted products to arrive at **the covariance**.

Con't....

$$\begin{aligned} \text{Cov} (R_i, R_j) = & p_1 [R_{i1} - E(R_i)] [R_{j1} - E(R_j)] \\ & \vdots \\ & + p_2 [R_{i2} - E(R_i)] [R_{j2} - E(R_j)] \\ & + \\ & + p_n [R_{in} - E(R_i)] [R_{jn} - E(R_j)] \end{aligned}$$

ILLUSTRATION

The returns on assets 1 and 2 under five possible states of nature are given below

State of nature	Probability	Return on asset 1	Return on asset 2
1	0.10	-10%	5%
2	0.30	15	12
3	0.30	18	19
4	0.20	22	15
5	0.10	27	12

The expected return on asset 1 is :

$$E(R_1) = 0.10 (-10\%) + 0.30 (15\%) + 0.30 (18\%) + 0.20 (22\%) + 0.10 (27\%) = 16\%$$

The expected return on asset 2 is :

$$E(R_2) = 0.10 (5\%) + 0.30 (12\%) + 0.30 (19\%) + 0.20 (15\%) + 0.10 (12\%) = 14\%$$

The covariance between the returns on assets 1 and 2 is calculated below :

<i>State of nature</i>	<i>Probability</i>	<i>Return on asset 1</i>	<i>Deviation of the return on asset 1 from its mean</i>	<i>Return on asset 2</i>	<i>Deviation of the return on asset 2 from its mean</i>	<i>Product of the deviations times probability</i>
(1)	(2)	(3)	(4)	(5)	(6)	(2) x (4) x (6)
1	0.10	-10%	-26%	5%	-9%	23.4
2	0.30	15%	-1%	12%	-2%	0.6
3	0.30	18%	2%	19%	5%	3.0
4	0.20	22%	6%	15%	1%	1.2
5	0.10	27%	11%	12%	-2%	-2.2
						Sum = 26.0

Thus the covariance between the returns on the two assets is 26.0.

Grouping Individual Assets into Portfolios

- The riskiness of a portfolio that is made of different risky assets is a function of three different factors:
 - the riskiness of the individual assets that make up the portfolio
 - the relative weights of the assets in the portfolio
 - the degree of **co-movement** of returns of the assets making up the portfolio.
- The **standard deviation** of a **two-asset portfolio** may be measured using the Markowitz model:

$$\sigma_p = \sqrt{\sigma_1^2 w_1^2 + \sigma_2^2 w_2^2 + 2w_1 w_2 \rho_{1,2} \sigma_1 \sigma_2}$$

Expected Return and Risk For Portfolios

Standard Deviation of a Two-Asset Portfolio using Covariance

[8-11]

$$\sigma_p = \sqrt{\underbrace{(w_A)^2(\sigma_A)^2}_{\text{Risk of Asset A adjusted for weight in the portfolio}} + \underbrace{(w_B)^2(\sigma_B)^2}_{\text{Risk of Asset B adjusted for weight in the portfolio}} + \underbrace{2(w_A)(w_B)(COV_{A,B})}_{\text{Factor to take into account comovement of returns. This factor can be negative.}}}$$

Risk of Asset A
adjusted for weight
in the portfolio

Risk of Asset B
adjusted for weight
in the portfolio

Factor to take into
account comovement of
returns. This factor
can be negative.

Example

- + Determine the expected return and standard deviation of the following portfolio consisting of two stocks that have a correlation coefficient of .75.

Portfolio	Weight	Expected Return	Standard Deviation
Apple	.50	.14	.20
Coca-Cola	.50	.14	.20

Answer

□ Expected Return of the Portfolio = $.5 (.14) + .5 (.14) =$ **.14 or 14%**

□ Standard deviation of the portfolio

$$= \sqrt{(.5^2 \times .2^2) + (.5^2 \times .2^2) + (2 \times .5 \times .5 \times .75 \times .2 \times .2)}$$
$$= \sqrt{.035} =$$
 .187 or 18.7%

➤ Lower than the weighted average of 20%.

PORTFOLIO RISK : 2 – SECURITY CASE

$$\sigma_p = [w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1w_2 \rho_{12} \sigma_1 \sigma_2]^{1/2}$$

Example : $w_1 = 0.6$, $w_2 = 0.4$,

$$\sigma_1 = 10\%, \sigma_2 = 16\%$$

$$\rho_{12} = 0.5$$

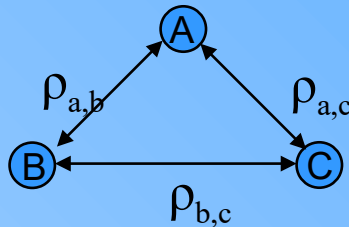
$$\begin{aligned}\sigma_p &= [0.6^2 \times 10^2 + 0.4^2 \times 16^2 + 2 \times 0.6 \times 0.4 \times 0.5 \times 10 \times 16]^{1/2} \\ &= 10.7\%\end{aligned}$$

■ The average standard deviation of two securities is 13, which is less than standard deviation of the portfolio, which is 10. Thus diversification reduces risk.

Risk of a Three-asset Portfolio

The data requirements for a three-asset portfolio grows dramatically if we are using Markowitz Portfolio selection formulae.

We need 3 (three) correlation coefficients between A and B; A and C; and B and C.



$$\sigma_p = \sqrt{\sigma_A^2 w_A^2 + \sigma_B^2 w_B^2 + \sigma_C^2 w_C^2 + 2w_A w_B \rho_{A,B} \sigma_A \sigma_B + 2w_B w_C \rho_{B,C} \sigma_B \sigma_C + 2w_A w_C \rho_{A,C} \sigma_A \sigma_C}$$

Three-Asset Portfolio Risk Matrix

	<i>Security X_1</i>	<i>Security X_2</i>	<i>Security X_3</i>
<i>Security X_1</i>	$X_1^2 \sigma_1^2$	$X_1 X_2 \rho_{12} \sigma_1 \sigma_2$	$X_1 X_3 \rho_{13} \sigma_1 \sigma_3$
<i>Security X_2</i>	$X_2 X_1 \rho_{21} \sigma_2 \sigma_1$	$X_2^2 \sigma_2^2$	$X_2 X_3 \rho_{23} \sigma_2 \sigma_3$
<i>Security X_3</i>	$X_3 X_1 \rho_{31} \sigma_3 \sigma_1$	$X_3 X_2 \rho_{32} \sigma_3 \sigma_2$	$X_3^2 \sigma_3^2$

P12 = correlation between security 1 and 2

P13 = correlation between security 1 and 3

P23 = correlation between security 2 and 3

PORTFOLIO RISK : 3 – SECURITY CASE

$$\sigma_p = [\sum \sum w_i w_j \rho_{ij} \sigma_i \sigma_j]^{1/2}$$

Example : $w_1 = 0.5$, $w_2 = 0.3$, and $w_3 = 0.2$

$$\sigma_1 = 10\%, \sigma_2 = 15\%, \sigma_3 = 20\%$$

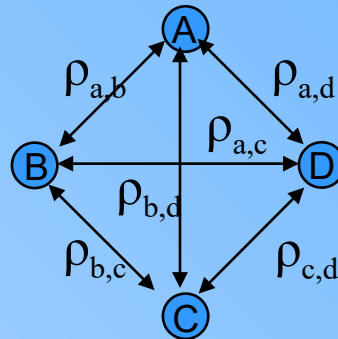
$$\rho_{12} = 0.3, \rho_{13} = 0.5, \rho_{23} = 0.6$$

$$\begin{aligned} \sigma_p &= [w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + w_3^2 \sigma_3^2 + 2 w_1 w_2 \rho_{12} \sigma_1 \sigma_2 \\ &\quad + 2w_2 w_3 \rho_{13} \sigma_1 \sigma_3 + 2w_2 w_3 \rho_{23} \sigma_2 \sigma_3]^{1/2} \\ &= [0.5^2 \times 10^2 + 0.3^2 \times 15^2 + 0.2^2 \times 20^2 \\ &\quad + 2 \times 0.5 \times 0.3 \times 0.3 \times 10 \times 15 \\ &\quad + 2 \times 0.5 \times 0.2 \times 0.5 \times 10 \times 20 \\ &\quad + 2 \times 0.3 \times 0.2 \times 0.6 \times 15 \times 20]^{1/2} \\ &= 10.79\% \end{aligned}$$

Risk of a Four-asset Portfolio

✓ The data requirements for a four-asset portfolio grows dramatically if we are using Markowitz Portfolio selection formulae.

✓ We need 6 correlation coefficients between A and B; A and C; A and D; B and C; C and D; and B and D.



Example

- Suppose Asset A has an expected return of 10 percent and a standard deviation of 20 percent. Asset B has an expected return of 16 percent and a standard deviation of 40 percent. If the correlation between A and B is 0.6, what are the expected return and standard deviation for a portfolio comprised of 30 percent Asset A and 70 percent Asset B?

Portfolio Expected Return

$$\begin{aligned}\hat{r}_P &= w_A \hat{r}_A + (1 - w_A) \hat{r}_B \\ &= 0.3(0.1) + 0.7(0.16) \\ &= 0.142 = 14.2\%.\end{aligned}$$

Portfolio Standard Deviation

$$\begin{aligned}\sigma_p &= \sqrt{W_A^2 \sigma_A^2 + (1 - W_A)^2 \sigma_B^2 + 2W_A(1 - W_A)\rho_{AB}\sigma_A\sigma_B} \\ &= \sqrt{0.3^2(0.2^2) + 0.7^2(0.4^2) + 2(0.3)(0.7)(0.4)(0.2)(0.4)} \\ &= 0.309\end{aligned}$$

RISK OF AN N - ASSET PORTFOLIO

$$\sigma_p^2 = \sum \sum w_i w_j \rho_{ij} \sigma_i \sigma_j$$

n x *n* MATRIX

	1	2	3	...	<i>n</i>
1	$w_1^2 \sigma_1^2$	$w_1 w_2 \rho_{12} \sigma_1 \sigma_2$	$w_1 w_3 \rho_{13} \sigma_1 \sigma_3$	...	$w_1 w_n \rho_{1n} \sigma_1 \sigma_n$
2	$w_2 w_1 \rho_{21} \sigma_2 \sigma_1$	$w_2^2 \sigma_2^2$	$w_2 w_3 \rho_{23} \sigma_2 \sigma_3$	...	$w_2 w_n \rho_{2n} \sigma_2 \sigma_n$
3	$w_3 w_1 \rho_{31} \sigma_3 \sigma_1$	$w_3 w_2 \rho_{32} \sigma_3 \sigma_2$	$w_3^2 \sigma_3^2$	...	
:	:				:
<i>n</i>	$w_n w_1 \rho_{n1} \sigma_n \sigma_1$				$w_n^2 \sigma_n^2$

Example 2

Suppose you have predicted the following returns on stock of X and Y company in four possible states of the economy as follows:

State of the economy	Probability	Return for stock X (Rx) (%)	Return for stock Y (Ry) (%)
Boom	0.25	18.5	15
Normal	0.25	10	11
Recession	0.25	2	6
Depression	0.25	-6	-4

Required:

Using the above data, calculate expected portfolio return and Risk if you invest 40% on stock X and 60% on Stock Y?

1. Find expected return of each security to find portfolio return

State	Pr.	(Rx) (%)	Ry (%)	PrRx	PrRy
Boom	0.25	18.5	15	4.625	3.75
Normal	0.25	10	11	2.5	2.75
Recession	0.25	2	6	0.5	1.5
Depression	0.25	-6	-4	-1.5	-1
sum				E(Rx)=6.13	E(Ry)=7

$$\begin{aligned} \text{➤ } E(R_x) &= 0.25(18.5) + 0.25(10) + 0.25(2) + 0.25(-6) \\ &= 6.13\% \end{aligned}$$

$$\text{➤ } E(R_y) = 0.25(15) + 0.25(11) + 0.25(6) + 0.25(-4) = 7\%$$

Cont....

2. Find standard deviation of each security

Pr.	R _x	R _y	(R _x -E(R _x))	(R _x -E(R _x)) ²	Pr (R _x -E(R _x)) ²	(R _y -E(R _y))	(R _y -E(R _y)) ²	Pr.(R _y -E(R _y)) ²
0.25	18.5	15	12.4	153.14	38.29	8	64	16
0.25	10	11	3.9	15.02	3.75	4	16	4
0.25	2	6	-4.13	17.02	4.254	-1	1	0.25
0.25	-6	-4	-12.125	147.02	36.754	-11	121	30.25
					var=83.045			Var=50.5

$$E(R_x)=6.13$$

$$E(R_y)=7$$

$$\sigma_x^2 = \sum p[R_x - E(R_x)]^2$$

$$\begin{aligned}\text{Var of } x = \sigma_x^2 &= 0.25(18.5-6.13)^2 + \\ & 0.25(10-6.13)^2 + \\ & 0.25(2-6.13)^2 + \\ & 0.25(-6-6.13)^2 \\ & = 38.2853 + 3.7539 + 4.25 + 36.75 = 83\end{aligned}$$

$$\text{St. D. of } x = \sigma_x = 9.11$$

$$\begin{aligned}\text{Var of } y = \sigma_y^2 &= 0.25(15-7)^2 + \\ & 0.25(11-7)^2 + \\ & 0.25(6-7)^2 + \\ & 0.25(-4-7)^2 \\ & = 16 + 4 + .25 + 30.25 = 50.5\end{aligned}$$

$$\sigma_y = 7.11$$

Cont....

3. Find covariance between X&Y

State	Pr.	(Rx)	Ry	Deviation (Rx-E(Rx))	Deviation (Ry-E(Ry))	Product of deviation (Rx-E(Rx)) (Ry-E(Ry))	P(Rx-E(Rx)) (Ry-E(Ry))
Boom	0.25	18.5	15	12.375	8	99	24.75
Normal	0.25	10	11	3.875	4	15.5	3.875
Recession	0.25	2	6	-4.125	-1	4.125	1.03125
Depression	0.25	-6	-4	-12.125	-11	133.375	33.34375
							Cov=63

$$Cov_{xy} = \sum \{P_i [R_x - E(R_x)] [R_y - E(R_y)]\}$$

$$Cov_{xy} = 0.25(18.5-6.13)(15-7.1) + 0.25(10-6.13)(11-7.1) + 0.25(2-6.13)(6-7.1) + 0.25(-6-6.13)(-4-7.1) =$$

$$= 24.75 + 3.875 + 1.03125 + 33.34375 = 63$$

$$R_{XY} = \frac{Cov_{xy}}{SD_x SD_y}$$

$$63 / (9.11 * 7.11) = 63 / 64.74 = 0.97$$

	Sec. X	Sec. Y
Exp. return	6.13	7
St. div.	9.11	7.11

Cont..

	Security x	Security y
Exp. return	6.13%	7%
St. D	9.11	7.11
Weight	40%	60%
Correlation = -0.97		

Portfolio variance

$$\begin{aligned}\sigma_p^2 &= w_x^2 \sigma_x^2 + w_y^2 \sigma_y^2 + 2w_x w_y \text{COV}_{xy} \\ &= w_x^2 \sigma_x^2 + w_y^2 \sigma_y^2 + 2w_x w_y \sigma_x \sigma_y \text{Cor}_{xy}\end{aligned}$$

$$\sigma_p^2 = (0.40^2)(9.11^2) + (0.60^2)(7.11^2) + 2(0.40)(0.60)(63) = 61.7$$

$$\sigma_p = 7.8\%$$

Or

$$\sigma_p^2 = (0.40^2)(9.11^2) + (0.60^2)(7.11^2) + 2(0.40)(0.60)(9.11)(7.11)(-0.97) = 61.7$$

**THANK U
AND
HAVE A NICE DAY!**